LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical and demonstration on Wires and cables: - Types of wires and cables, routing of wires and cables, open loom, ducted loom conduits, cable loom, and co-axial cables. Special purpose cables: - Ignition cables, thermocouple cables.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical and demonstration on Wires and cables: - Types of wires and cables, routing of wires and cables, open loom, ducted loom conduits, cable loom, and co-axial cables. Special purpose cables: - Ignition cables, thermocouple cables.
5. Specific Objectives	13.1.2.1 Knowledge of Static converting equipment, Bus bars, cable terminations, Switches.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of wire types, routing methods, open loom, ducted loom, cable loom, co-axial cables, ignition cables, and thermocouple cables. Safety: Wear PPE, verify power off. Instructor supervision: Monitor handling and routing techniques.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a cable loom?
(b) Define the difference between open loom and ducted loom.
(c) List two types of special purpose cables mentioned in the syllabus.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of wires and cables • Routing of wires and cables • Open loom and ducted loom conduits • Cable loom construction • Co-axial cable characteristics	L & Ds	PPT, IB	15 min
(b)	• Special purpose cables: Ignition cables • Special purpose cables: Thermocouple cables • Comparison of standard vs special cables • Inspection criteria for cable integrity • Routing constraints for high-voltage cables	L & Ds	PPT, PDF	20 min
(c)	• Summary of cable termination points • Bus bar connection basics • Switch integration with cabling • Static converting equipment overview • Final routing verification checklist	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why ignition cables require specific routing compared to standard wires.

(b) Describe the advantages of using a ducted loom over an open loom in a high-vibration environment.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between cable loom and open loom routing, referencing the next lesson on earthing connections.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Earthing or grounding connections, cable terminations, crimped terminals and Its advantages. Aluminum cable connections, Plugs and sockets. Different types of plugs and sockets potting, need for electrical Bonding. Static discharge wicks. Bonding and screening methods. Practical on crimping, bonding screening, earthing & grounding.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min